

PILLAR 02 // DUAL-AIRFRAME DEFENCE ECOSYSTEM

MODULAR AIRFRAME ARCHITECTURE & SPECIFICATIONS

Dual-Platform Capability: High-Endurance VTOL Hybrid and Precision Heavy-Lift Multirotor



PLATFORM 01 • SHIV-NETRA47 VTOL HYBRID

ENDURANCE LOGISTICS & ISR

180 MIN
ENDURANCE

150 KM
RADIO RANGE

5,500 M
CEILING

200 KG
MTOW TARGET

Twin-boom aerodynamic carbon airframe combining zero-runway quad-rotor vertical takeoff with high-speed fixed-wing pusher cruise. Optimized for high-altitude mountain corridor resupply and over-the-horizon border surveillance.

PLATFORM 02 • SHIV-NETRA47 TACTICAL MULTIROTOR

PRECISION TACTICAL CARGO

45 MIN
HOVER TIME

25 KG
USEFUL PAYLOAD

4,500 M
CEILING

6 ROTOR
REDUNDANCY

Heavy-lift hexacopter with folding carbon arms designed for pinpoint vertical delivery into restricted mountain trenches, artillery munitions airlift, winch deployment, and low-altitude high-resolution 3D LIDAR terrain scanning.

6 SWAPPABLE LINE-REPLACEABLE UNIT (LRU) SUBSYSTEMS

MODULAR DEFENCE ARCHITECTURE

LRU 01
AIRFRAME CORE

Full pre-preg carbon fiber monocoque fuselage with IP67 all-weather sealing.

LRU 02
TRIPLE AUTOPILOT

Triple-redundant IMU avionics with dual NAVIC/GPS and optical flow fallover.

LRU 03
PROPULSION

Brushless vertical lift motors + heavy pusher cruise motor with FOC telemetry.

LRU 04
TACTICAL COMMS

AES-256 encrypted FHSS datalink with MANET swarm mesh relay capability.

LRU 05
CARGO BAY

Quick-swap bottom rail for medical cold boxes, munition pods, or EO/IR cameras.

LRU 06
RUGGED GCS

Mk-STD-810H dual-display portable command station with hands-on controls.

FORMAL ENGINEERING SPECIFICATION BENCHMARK		CERTIFICATION STANDARD: EASA SAIL III / SORA 2.5 • DOCA TYPE CERTIFICATION READY
TECHNICAL SPECIFICATION	SHIV-NETRA47 VTOL HYBRID (LONG RANGE)	SHIV-NETRA47 TACTICAL MULTIROTOR (HEAVY LIFT)
Max Takeoff Weight (MTOW)	45 kg (scalable up to 200 kg class target)	55 kg (heavy-lift tactical multirotor)
Useful Payload Capacity	5 – 15 kg (internal modular hermetic cargo pod)	15 – 25 kg (external underslung or winch payload)
Maximum Flight Endurance	Up to 180 Minutes (fixed-wing aerodynamic pusher cruise)	45 – 55 Minutes (continuous precision hovering)
Operational Command Range	150 km Line-of-Sight / Unlimited via MANET Swarm Relays	25 km Tactical Perimeter Operations
Maximum Service Ceiling	5,500 m AMSL (Himalayan passes and ridge outposts)	4,500 m AMSL (valley floor & forward trenches)
Cruise / Max Dash Speed	105 km/h cruise / 145 km/h dash speed	55 km/h transit / 0 km/h precision hover
Propulsion Architecture	Hybrid: 4x VTOL vertical lift rotors + 1x rear pusher prop	Hexacopter: 6x heavy-lift coaxial/direct-drive rotors
European Flight Heritage	Flight-proven CATUAV / BCN Drone Center European airframe	Flight-tested industrial heavy-lift multirotor platform

SHIV-NETRA47 AIR DEFENCE SYSTEM PRIVATE LIMITED
CIN: U30302UP2024PTC202547 • Make in India Defence UAS & Aerial Systems

Technology Support: BCN Drone Center (Moià, Spain) • Web: www.shivnetra47.com

PAGE 03 / 05